Spectral Overlap and Structural Degeneracy in Atmospheric Retrieval Architectures: A Controlled Synthetic Experiment

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Abstract

Atmospheric retrieval techniques are widely used to infer molecular composition from exoplanet transmission spectra. However, the relationship between observed spectra and inferred molecular abundances can depend strongly on the structure of the retrieval architecture and the spectral similarity between molecular bases.

In this work we present a controlled synthetic experiment designed to quantify how spectral overlap between molecular templates affects the stability and interpretability of atmospheric retrieval solutions. Starting from a synthetic spectrum generated with H_2O, CH_4, and a cloud component, we construct alternative molecular bases with tunable spectral similarity to the true CH_4 basis. By continuously varying the overlap between the true and alternative molecular bases, we map the resulting statistical penalties and spurious molecular attributions across retrieval solutions.

The results reveal a continuous degeneracy regime: as spectral overlap increases, the statistical distinguishability between correct and incorrect retrieval architectures decreases, while stable but spurious molecular attributions persist. These findings highlight the existence of controlled structural degeneracy regimes in atmospheric retrieval and illustrate how inference outcomes can depend on the architectural structure of the retrieval basis.

1 Introduction

Atmospheric retrieval has become a central tool in the interpretation of exoplanet transmission spectra. By fitting parametric models of atmospheric composition and cloud structure to observed spectra, retrieval frameworks aim to infer atmospheric molecular abundances from spectroscopic observations.

However, the inference process depends not only on the observed spectrum but also on the set of molecular bases included in the retrieval architecture. When spectral features from different molecular species exhibit partial overlap, the retrieval problem can become degenerate: different molecular bases may produce similar spectral signatures within the noise level of the observations.

Such degeneracies are often discussed qualitatively in the literature, but their structural dependence on spectral overlap is rarely explored in a controlled setting. In particular, it remains useful to understand how inference behavior changes continuously as the similarity between molecular templates varies.

In this work we construct a synthetic experiment that allows direct control over the spectral overlap between molecular bases. By introducing an alternative molecular basis with tunable similarity to the

true molecular template, we systematically explore how atmospheric retrieval behavior transitions from identifiable to quasi-degenerate regimes.

The goal of this work is not to reproduce a specific observational dataset but rather to illustrate the structural behavior of retrieval architectures under controlled conditions.

2 Synthetic Experiment

2.1 Synthetic spectrum

We generate a synthetic transmission spectrum over the wavelength range 1–5 μm. The spectrum is constructed as a combination of:

- a smooth continuum component
- three H_2O absorption features
- three CH_4 absorption features
- a simple cloud masking term

Gaussian noise is added to simulate observational uncertainty.

The resulting synthetic spectrum therefore represents a simplified but controlled atmospheric signal with known molecular components.

2.2 Retrieval architectures

Two retrieval architectures are compared.

Retrieval B (reference architecture)

H_2O + CH_4 + clouds

Retrieval A (alternative architecture)

H_2O + ALT + clouds

where **ALT** is an alternative molecular basis constructed to approximate the CH_4 spectral structure with controlled distortions.

The alternative molecular basis is defined as

$$ALT = CH_4 + \alpha \times distortion$$

where α controls the degree of distortion relative to the true CH_4 template. Increasing α decreases the spectral similarity between the alternative basis and the true CH_4 basis.

Each retrieval model is fit using weighted least squares to the synthetic spectrum.

3 Controlled Spectral Overlap Scan

To quantify the effect of spectral similarity on retrieval outcomes, we perform a scan over the distortion parameter α.

For each value of α we compute:

- the mean spectral overlap between the CH_4 and ALT bases
- the mean χ^2 value for Retrieval A
- the mean χ^2 value for Retrieval B
- the resulting statistical penalty

$\Delta\chi^2 = \chi^2(A) - \chi^2(B)$

Each configuration is evaluated across multiple noise realizations in order to estimate stable mean values.

The spectral overlap between molecular bases is measured as the mean cosine similarity between the corresponding spectral components.

4 Results

4.1 Degeneracy phase diagram

The controlled overlap scan produces a continuous degeneracy phase diagram relating:

- spectral overlap between molecular bases
- statistical distinguishability of retrieval architectures
- strength of alternative molecular attribution

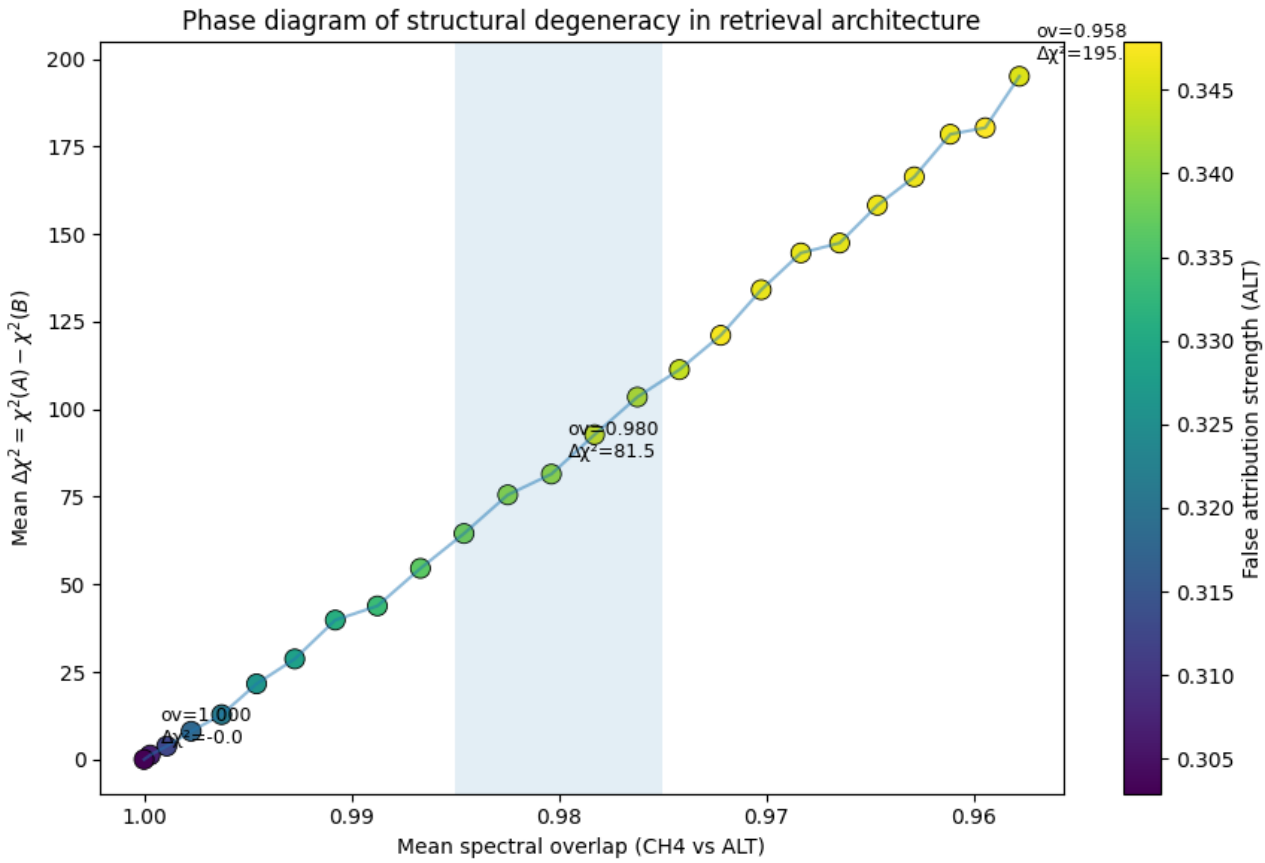

Figure 1 — Degeneracy phase diagram.
Mean statistical penalty $\Delta\chi^2$ between retrieval architectures as a function of spectral overlap between the CH_4 and alternative molecular bases. Increasing overlap reduces the statistical distinguishability between the models and defines a quasi-degenerate regime.

As spectral overlap approaches unity, the statistical distinguishability between the architectures decreases continuously.

For example, at overlap ≈ 0.98 we obtain

$\Delta\chi^2 \approx 81$ (averaged over noise realizations)

indicating that the alternative architecture remains statistically penalized but not catastrophically inconsistent with the data.

This intermediate region defines a quasi-degenerate regime in which alternative molecular interpretations remain statistically plausible despite being structurally incorrect.

4.2 Stability of spurious molecular attribution

Across the overlap scan we observe that the retrieved strength of the alternative molecular component remains relatively stable, even as the statistical penalty varies substantially.

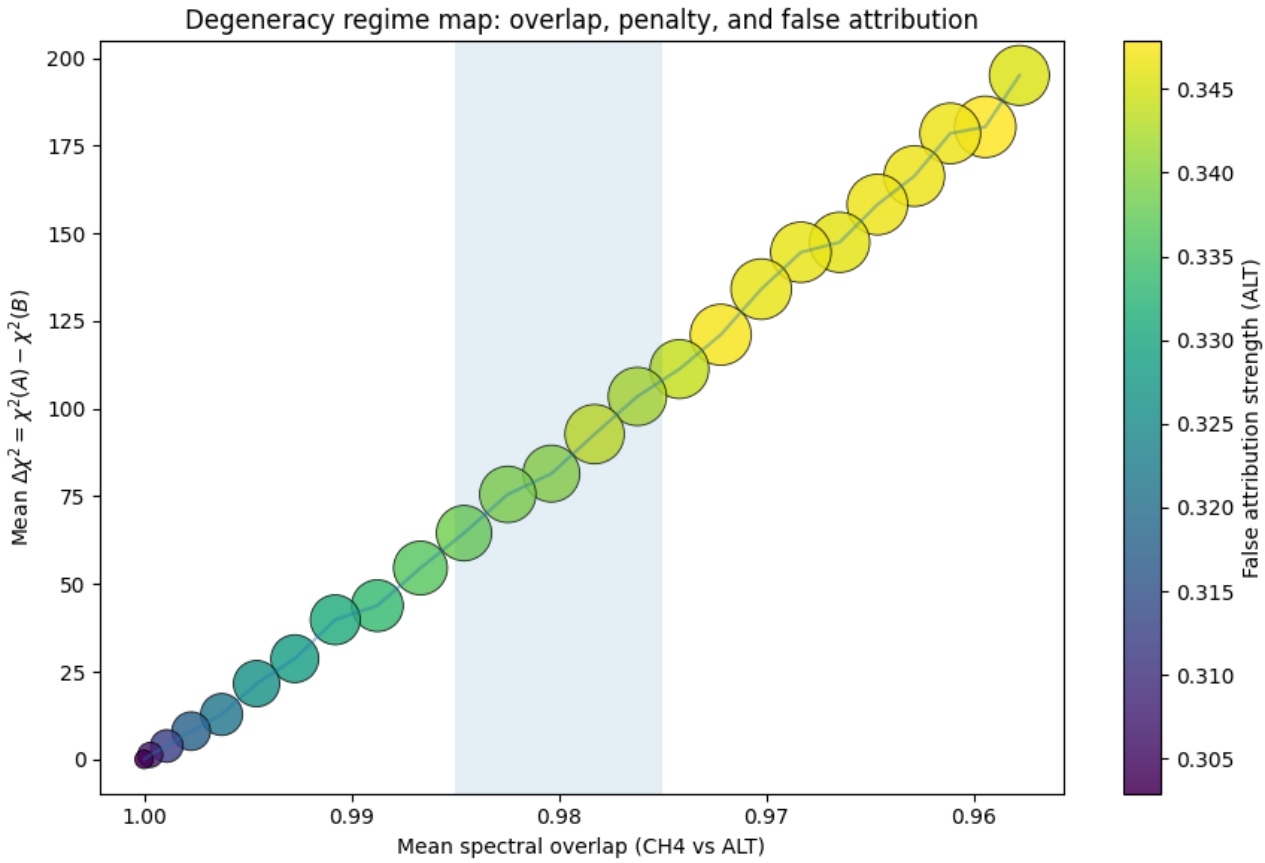

Figure 2 — Degeneracy regime map.
Joint representation of spectral overlap, statistical penalty, and spurious molecular attribution strength across the overlap scan. Bubble size represents the inferred strength of the alternative molecular component.

This indicates that retrieval architectures can produce stable but spurious molecular attributions when the spectral bases exhibit sufficient similarity.

Such behavior arises naturally from the structure of the inference problem rather than from numerical instability.

4.3 Representative example

A representative configuration occurs at spectral overlap ≈ 0.98.

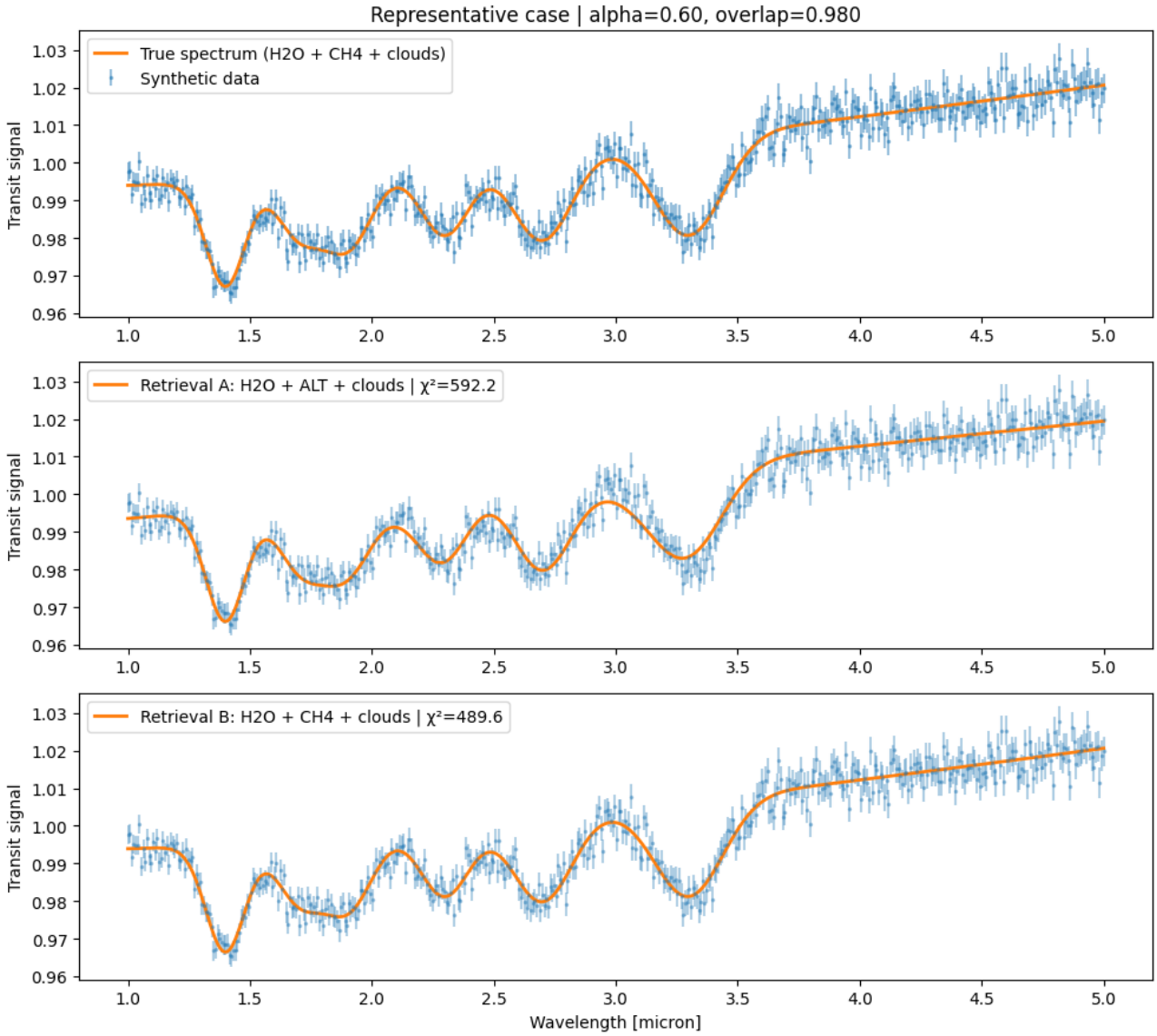

Figure 3 — Representative quasi-degenerate retrieval case.
Comparison between the true synthetic spectrum and the fits produced by the alternative and reference retrieval architectures. The alternative architecture reproduces the overall spectral shape but produces systematic residuals relative to the reference model.

In this case the alternative retrieval architecture reproduces the overall spectral shape but produces systematic residuals relative to the reference model.

This illustrates the characteristic behavior of quasi-degenerate retrieval solutions: visually plausible spectral fits that nevertheless incur measurable statistical penalties.

5 Structural robustness of the degeneracy regime

To verify that the observed degeneracy behavior is not specific to the particular synthetic spectrum used in the experiment, we repeat the overlap scan under multiple random noise realizations and alternative distortion patterns applied to the CH_4 template.

Across these variations, the overall structure of the degeneracy phase diagram remains stable: increasing spectral overlap consistently reduces the statistical distinguishability between retrieval architectures while maintaining persistent alternative molecular attribution.

This indicates that the observed degeneracy regime arises from the structural similarity between molecular bases rather than from specific features of the synthetic spectrum.

6 Discussion

The synthetic experiment presented here highlights a structural feature of atmospheric retrieval architectures: the mapping from spectral observations to inferred molecular composition is not uniquely determined by the spectrum alone but depends on the structure of the molecular basis used in the retrieval.

When molecular templates exhibit sufficient spectral similarity, retrieval architectures can produce stable alternative explanations of the same spectrum.

The resulting degeneracy regimes are controlled by the spectral overlap between molecular bases and can therefore be explored systematically through controlled synthetic experiments.

These results do not imply that molecular detections from atmospheric retrieval are unreliable. Rather, they illustrate the importance of carefully considering architectural assumptions when interpreting retrieval outcomes.

7 Conclusion

We presented a controlled synthetic experiment exploring how spectral overlap between molecular templates influences atmospheric retrieval solutions.

By introducing alternative molecular bases with tunable similarity to the true molecular template, we constructed a degeneracy phase diagram linking spectral overlap to statistical distinguishability between retrieval architectures.

The results show that increasing spectral similarity leads to a continuous transition from identifiable solutions to quasi-degenerate regimes where alternative molecular attributions remain stable but statistically penalized.

These findings illustrate how inference outcomes can depend on the structure of the retrieval architecture and highlight the usefulness of controlled synthetic experiments for exploring structural degeneracy in atmospheric retrieval.

Code availability

The Python notebook used to generate the synthetic experiment
and figures is included in the record.